

Name			
USN			
Date	Conduction	Record	Viva
Allotted Marks			
Obtained Marks			

EXPERIMENT 1

Install Hadoop and Implement the following file management tasks in Hadoop: Adding files and directories Retrieving files Deleting files and directories.

Aim: To Install Hadoop and Implement the following file management tasks in Hadoop:
Adding files and directories Retrieving files Deleting files and directories

Introduction:

Every business is now a data business. Data is the organizations' future and most valuable asset. The Hortonworks Data Platform (HDP) is a security-rich, enterprise-ready open-source Hadoop distribution based on a centralized architecture (YARN). Hortonworks Sandbox is a single-node cluster and can be run as a Docker container installed on a virtual machine. [HDP](#) is a complete system to handle the processing and storage of big data. It is an open architecture used to store and process complex and large-scale data. It is composed of numerous Apache Software Foundation (ASF) projects including Apache Hadoop and is built specifically to meet enterprise demands. Hortonworks was a standalone company until 2019 when it is merged to Cloudera and now Hortonworks is a subsidiary for Cloudera, Inc.



CLUDERA

Hortonworks is merged to Cloudera in 2019

Hardware requirements

- **Memory** dedicated to the cluster **Recommended: 16+ GiB**, More is better.
- **CPU** (Minimum: 4 Cores, **Recommended: 6+ Cores**)
 - Virtualization should be enabled

Storage

- 65-75 GiB
 - for HDP 2.6.5

Installing Hortonworks Sandbox

There are two common ways to install HDP Sandbox on your PC, either by using a hypervisor such as *VirtualBox* which will pull the Docker image and run a container for your cluster or by directly using Docker where you need to manage your resources via docker command line [options](#) in Linux or you can use WSL backend in Windows where you can configure resources via .wslconfig file.

Installing a Hypervisor

follow the attached link in the following list to download your preferred hypervisor.

- [Oracle VM VirtualBox \(Recommended\)](#)
- [VMware Workstation Player](#) (Only for Linux and Windows)
- [VMware Fusion for mac](#) (Only for macOS)

Installing HDP Sandbox

Hortonworks Data Platform (HDP) is a straightforward, pre-configured, learning environment that contains the latest developments from Apache Hadoop. The Sandbox comes packaged in a virtual environment that can run in the cloud or on your personal machine. The Sandbox allows you to learn and explore HDP on your own.

Installing sandbox

The download links of HDP Sandbox on VirtualBox:

https://archive.cloudera.com/hwx-sandbox/hdp/hdp-2.6.5/HDP_2.6.5_virtualbox_180626.ova
<https://download.virtualbox.org/virtualbox/7.1.6/VirtualBox-7.1.6-167084-Win.exe>

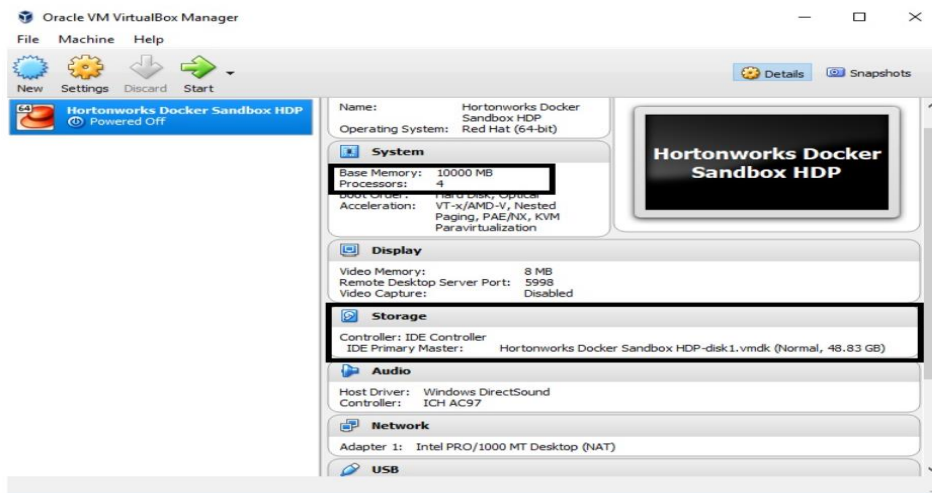
Install the HDP Sandbox on VirtualBox.

Once the installation is complete install PuTTY using the below link

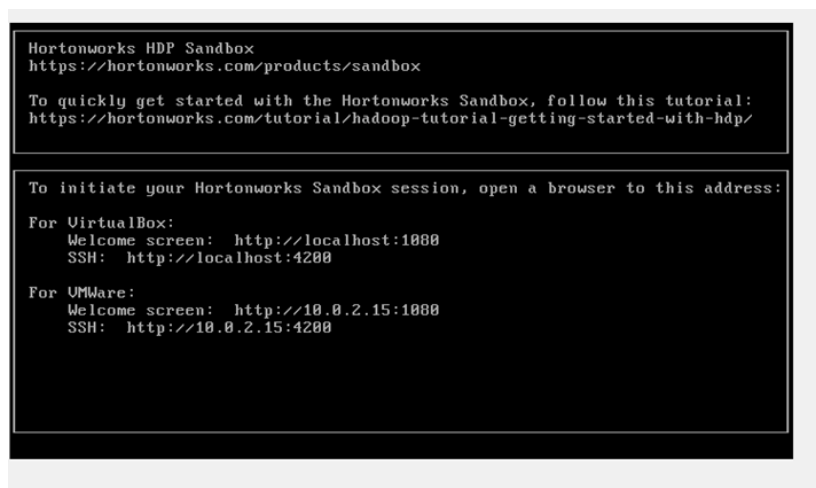
<https://the.earth.li/~sgtatham/putty/latest/w64/putty-64bit-0.83-installer.msi>

File management tasks in Hadoop

Open Virtual toolbox

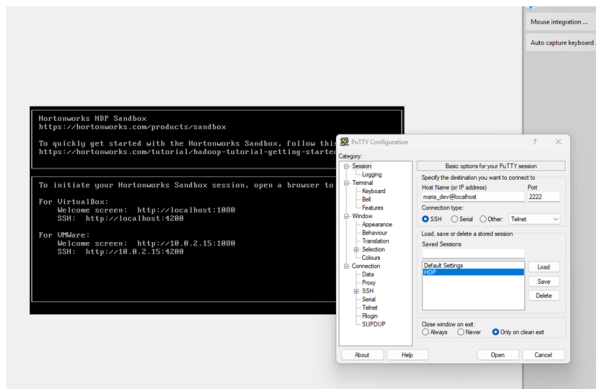


Press start



You will get the above screen.

Then click putty, you get the below screen



In the PuTTY configuration

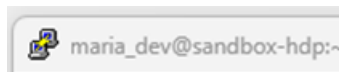
In the Host Name for IP address

Type: **maria_dev@localhost** & Port:2222

In the saved sessions

Select HDP & press Save , as you do so, HDP appears in the list then select load & open

You get a PuTTY cmp screen with following



Using username "maria_dev".

maria_dev@localhost's password:

Last login: Fri Mar 14 10:10:06 2025 from 172.18.0.3

It asks for password: type **maria_dev**

➤ To create a directory

➤ **To add the file to directory**

➤ **To delete a file**

➤ **To delete a directory**

.

Inference: